

Sambina Islam Aninta

PhD in AI x Genomics

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PROFESSIONAL SUMMARY

PhD student in computational biology at the University of British Columbia, advised by Dr. Carl de Boer. Research centers on understanding how deep learning models learn to predict functional consequences of genetic variants. Core work spans developing deep learning models for various functional genomics assays, constructing benchmarking framework for post-hoc interpretability methods for making these predictions interpretable and grounded in biological mechanisms.

CORE EXPERTISE

- Deep Learning for Regulatory Genomics
- Interpretable AI
- Functional Genomics

RESEARCH EXPERIENCE

University of British Columbia

Vancouver, BC

PhD Student · School of Biomedical Engineering

2025 – Present

Research Assistant/ Research Technician III

2024 – 2025

- Advisor – Dr. Carl de Boer, Supervisory Committee – Dr. Peter Koo, Dr. Ali Bashashati, Dr. Xin Tang, Dr. Sheila Teves
- Investigated how assay-specific biases influence predictive modeling in genomics, with applications in prioritizing the effects of genetic variants
- Establishing a benchmarking framework for evaluating post hoc interpretability methods applied to genomics deep learning models
- Developing generative models to study chromatin accessibility and DNA methylation using functional genomics

University of British Columbia

Vancouver, BC

Honors Thesis · School of Biomedical Engineering

2022 – 2023

- Advisor – Dr. Roger Tam
- Led an independent honors thesis research project developing and evaluating supervised machine learning models to predict exacerbation severity in patients with chronic obstructive pulmonary disease (COPD)

Amgen

Vancouver, BC

Data Informatics/Molecular Biology Co-op

2021 – 2021

- Developed and tested novel methodologies to support in vitro and in vivo human antibody discovery platforms
- Performed laboratory tasks such as RNA extraction, RNA purification, cDNA synthesis and gene amplification from a variety of sources such as single B cells, cultured B cells and hybridomas
- Designed and built a Dash web app to visualize immune repertoire containing thousands of sequences and deployed it in Kubernetes that increased efficiency of sequence discovery platforms

Michael Smith Genome Sciences Centre

Vancouver, BC

Bioinformatics Co-op · BC Cancer

2020 – 2021

- Advisor – Dr. Inanc Birol
- Co-developed an *in silico* pipeline to discover novel antimicrobial peptides (AMPs) from amphibian and insect transcriptomic data that can work as antibiotic alternatives

EDUCATION

Ph.D., Biomedical Engineering, University of British Columbia
B.Sc., Bachelor of Science in Combined Honours Computer Science and Biology,
University of British Columbia

2025 – 2029

2018 – 2023

PUBLICATIONS

Genomics X AI

- Aninta, ..., de Boer. Position-dependent variant effects reveal importance of context in genomic regulation. *bioRxiv*. 2026

Antimicrobial Peptides (AMP)

- ..., Aninta, ..., Birol. Mining the UniProtKB/Swiss-Prot database for antimicrobial peptides. *Protein Science*. 2025
- ..., Aninta, ..., Birol. Mining Amphibian and Insect Transcriptomes for Antimicrobial Peptide Sequences with rAMPage. *Antibiotics*. 2022

TALKS

2026

- Aninta, ..., de Boer. Position-dependent variant effects reveal importance of context in genomic regulation. *Intelligent Systems for Molecular Biology (ISMB)*

POSTERS

2026

- Aninta, ..., de Boer. Position-dependent variant effects reveal importance of context in genomic regulation. *Cold Spring Harbor Laboratory (CSHL) 90th Symposium AI in Bio*

2025

- Aninta, ..., de Boer. Variant Position Modulates Functional Impact in Massively Parallel Reporter Assays. *Biology of Genomes, Cold Spring Harbor Laboratory (CSHL)*

2024

- Aninta, ..., de Boer. How does data leakage from genomics foundational models inflate downstream model performance? *Biological Data Science, Cold Spring Harbor Laboratory (CSHL)*
- Aninta, ..., de Boer. How much of transfer learning is actually cheating? *MLCB*
- Aninta, ..., de Boer. Predicting variant effects using multimodal deep neural networks in autoimmune diseases. *SynBio6.0*

AWARDS

Title	Dates	Amount
SBME Travel Award	2026	\$1500
ISMB Travel Fellowship	2026	\$1,047
BIPOC Graduate Excellence Award	2025 – 2026	\$1,500
UBC Four Year Doctoral Fellowship (4YF)	2026 – 2030	\$72,000
President's Academic Excellence Initiative PhD Award	2025 – 2026	\$1,000
KM International Leader of Tomorrow (KMILOT) Award	2018 – 2023	\$250,000
UBC Trek Excellence Scholarship	2019 – 2020; 2022 – 2023	\$6,000
UBC Science Scholar	2018 – 2019, 2021 – 2023	
UBC Dean's Honour List	2018 – 2023	
UBC J Fred Muir Memorial Scholarship in Science	2019 – 2020	\$350
UBC Clayton Person Memorial Scholarship in Biology	2021 – 2022	\$750
UBC Charles and Jane Banks Scholarship	2022 – 2023	\$270
UBC Faculty of Science International Student Scholarship	2021 – 2023	\$7,700

REVIEWING

Nature, PLOS Computational Biology, Cell Genomics, ISMB

2025 – Present

LEADERSHIP

BC Director of Communications - Nucleate Canada

2024 – 2025

President, UBC STEM Fellowship

2021 – 2023

MENTORING & TEACHING

UBC Academic Essentials AI Coordinator

2026

Teaching Assistant for Applied Machine Learning (CPSC 330)

2023

AMS Tutoring Supervisor

2018 – 2023